

B9AI105 · REINFORCEMENT LEARNING · CA1 DEFENCE



GréineGrid Qagent

Tabular reinforcement learning for home battery dispatch
on a solar-connected microgrid

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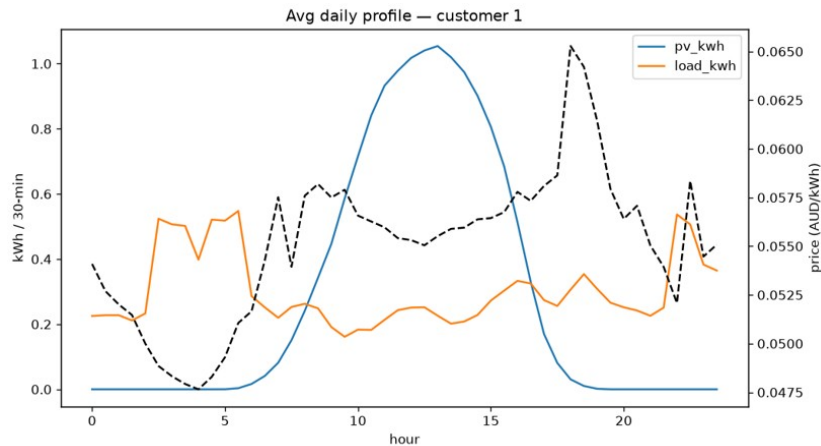
Live demo greineq-agent.sudocod.com

Solar peaks at noon. Demand peaks at night.

Without storage, surplus midday solar is exported at a low feed-in rate while the evening peak is bought at retail prices.

A home battery can shift energy in time — but every 30 minutes it must decide: hold, charge, or discharge.

48 decisions per day. No labelled “correct” action exists.



Avg daily profile — Ausgrid Customer 1 (load, PV, NSW1 price)

Real data: Ausgrid half-hourly household load + rooftop PV (2012–13) • AEMO NSW1 wholesale spot prices, merged per 30-min interval

Method, in three numbers

324

discretised states

SOC $\times$ PV $\times$ load $\times$ price $\times$ time-of-day
(tertile bins, fitted on train only)

2

agents from scratch

Q-Learning (off-policy) & SARSA (on-policy) — NumPy only, no RL libraries

4

baselines

no-battery floor $\cdot$ tertile rule $\cdot$ greedy
self-consumption $\cdot$ perfect-foresight
oracle

MDP: 48-step daily episodes $\cdot$ 3 actions (hold / charge from surplus / discharge to deficit) $\cdot$ 10 kWh, 5 kW battery $\cdot$ $\gamma = 0.95$ $\cdot$ 10,000 episodes $\cdot$ greedy test evaluation ($\epsilon = 0$) on 53 held-out days

Reward design decided what the agent learned

v1 — wholesale proxy

- Penalised spot price only — no retail margin, no feed-in
- Agent beat the rule baseline on reward...
- **...while losing on the actual bill (23.02 vs 18.80 AUD)**

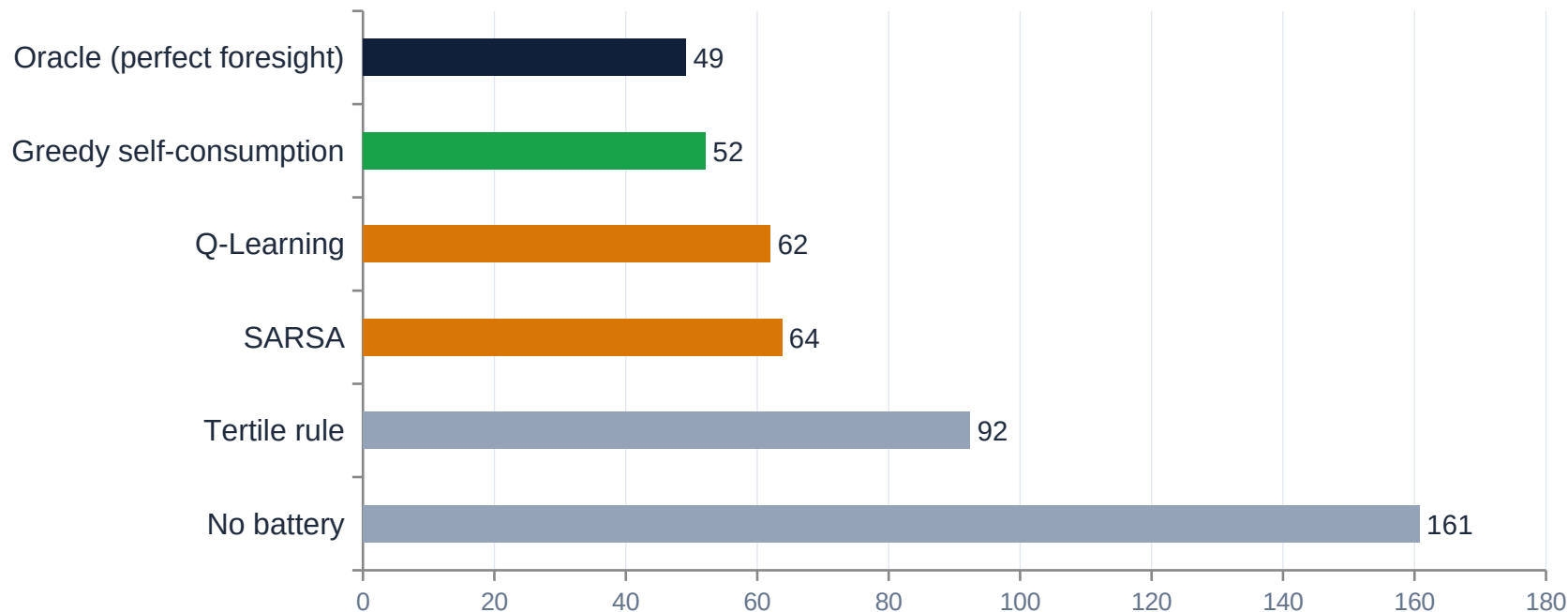


v2 — monetised AUD cost

- $r = -(\text{import} \times \text{retail}) - (\text{waste} \times \text{feed-in}) - \text{degradation}$
- Training objective = evaluation metric (household bill)
- **Reward ranking now matches bill ranking**

“The training reward diverged from the evaluation metric — the agent could look better on paper while doing worse on the bill.”

Total test-split cost, 53 days — lower is better



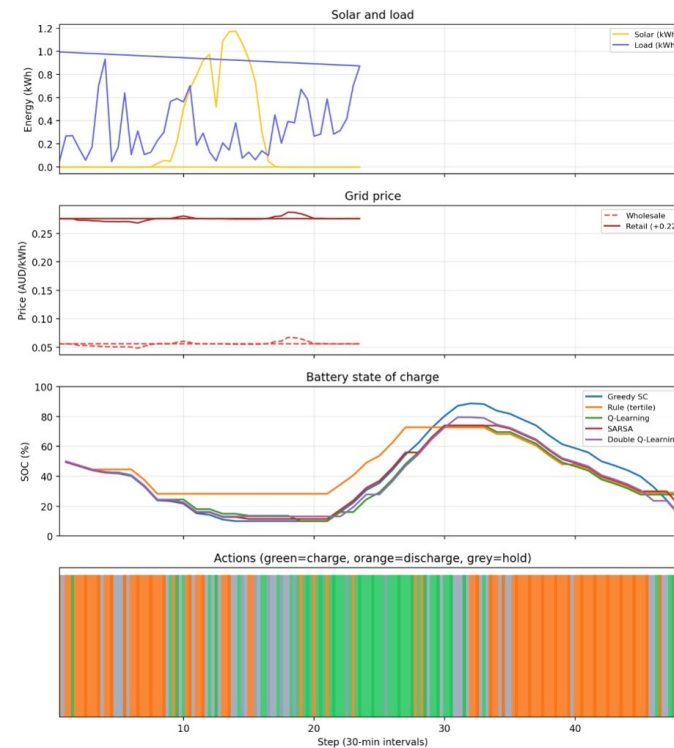
Q-Learning -32.9% and SARSA -31.0% vs the tertile rule — but both sit 19-22% above greedy self-consumption.

The headline finding

97.4%

of the oracle's achievable saving is already captured by simple greedy self-consumption

- Self-consumption rate: 0.76 greedy vs 0.60 rule
- Self-sufficiency: 0.80 greedy vs 0.58 rule
- Evening peak import: 0.77 vs 1.74 kWh/step
- **Bottleneck = problem structure + state coarseness, not the TD algorithm**



Digital twin — multi-policy replay of one test day

What we're not claiming



Known limitations

- Single customer; 2012–13 tariffs
- Solar-only dispatch — no grid charging or export arbitrage
- Time-of-day bins break strict Markov property
- **5 seeds × 10k: Q-Learning 61.75 ± 2.14 AUD, SARSA 61.62 ± 0.86 AUD (paired t-test $p = 0.90$)**

Where RL earns its place

Correct wiring + a simple controller already delivers most of the benefit here.

RL's value rises with grid charging, price arbitrage, and fleet-level coordination — scoped as CA2 extensions on this same MDP and codebase.

Not because RL failed here — because the problem hasn't needed it yet.



Conclusion

- Reward design determines what the agent learns — $v1 \rightarrow v2$ fixed a real training/evaluation misalignment
- Q-Learning & SARSA beat weak rules (-31-33%), not strong heuristics (+19-22% vs greedy SC)
- Greedy self-consumption sets a high, realistic ceiling — 97.4% of achievable savings
- Next (CA2): grid charging, tariff arbitrage, cross-household transfer

Questions? • greineq-agent.sudocod.com